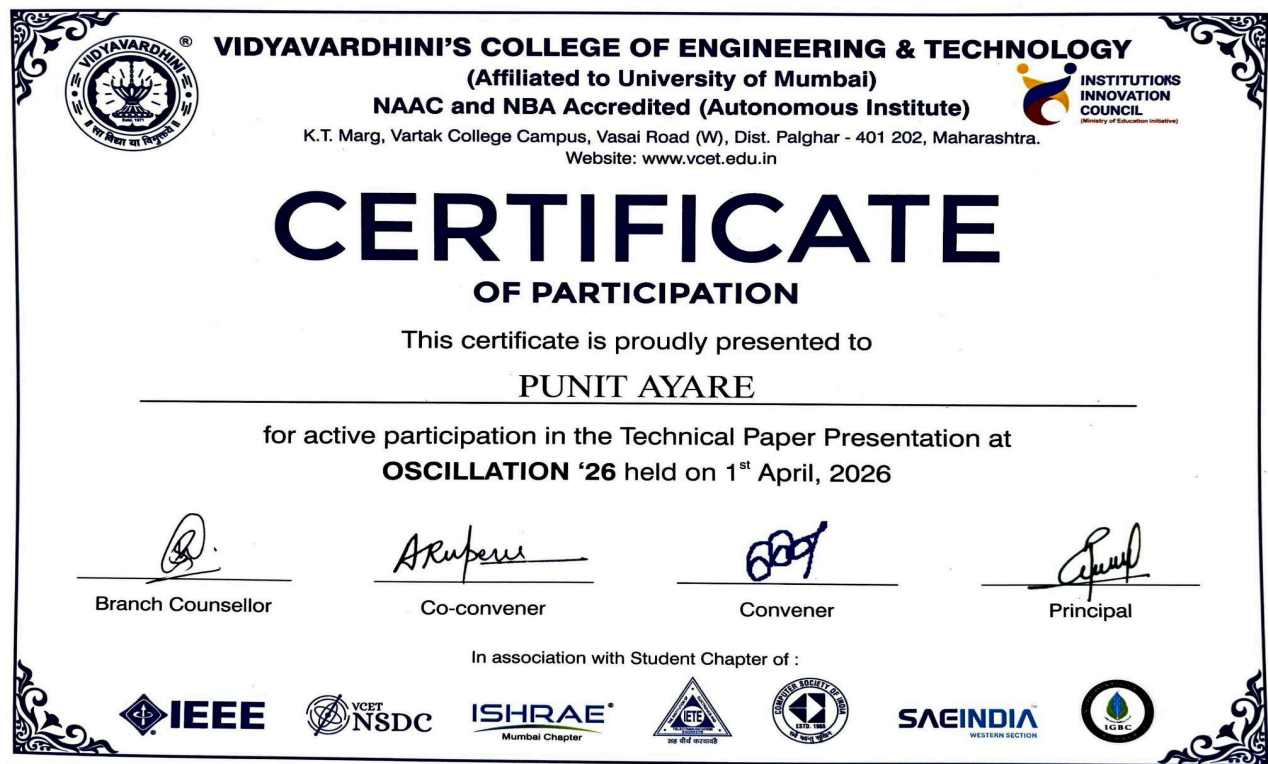


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PLAGIARISM REPORT



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Plagiarism Scan Report



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Abstract

This project presents FallAlert, an integrated AI-powered system designed for real-time fall detection and emergency notification to assist the elderly and high-risk patients. Utilizing an MPU6050 6-axis sensor interfaced with an ESP32 microcontroller, the system captures complex human motion data including linear acceleration and angular velocity. A machine learning model, trained on diverse movement datasets, analyzes this data with an accuracy of 85-95% to effectively distinguish between activities of daily living and actual fall events. Upon detection, the system triggers instant push notifications via Firebase Cloud Messaging (FCM) to a dedicated Flutter mobile application used by caregivers. This end-to-end solution ensures rapid response times, significantly reducing the "long lie" period and potentially saving lives by accelerating medical intervention

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List of Abbreviations

AI – Artificial Intelligence

IoT – Internet of Things

ESP32 – Espressif Systems 32-bit Microcontroller

MPU6050 – Motion Processing Unit with Accelerometer and Gyroscope API – Application

Programming Interface

FCM – Firebase Cloud Messaging

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CHAPTER NO. 1	
INTRODUCTION	

1.1 Overview

The FallAlert system is a comprehensive safety solution designed to bridge the gap between wearable hardware and immediate emergency response. As the global elderly population increases, the risk of unmonitored falls leading to severe injury or death has become a critical healthcare concern. Our system leverages the ESP32 microcontroller for its low power consumption and built-in Wi-Fi capabilities, allowing for seamless data transmission to the cloud. By combining Internet of Things (IoT) hardware with an AI-driven backend, the project provides a reliable, 24/7 monitoring net that functions without the need for manual user intervention.

1.2 Problem Statements & Objectives

Problem Statement: Delayed medical assistance following a fall is a leading cause of long-term complications in the elderly. Manual alert systems are often unusable if the patient is unconscious or in shock.

Objectives:

- To develop a wearable device using MPU6050 for continuous motion monitoring.
- To implement a Machine Learning model for high-accuracy fall detection.
- To provide caregivers with real-time alerts through a mobile application.
- To maintain a digital history of fall events for clinical analysis.

1.3 Scope

The scope of this project encompasses the development of the ESP32 firmware for data acquisition and the deployment of a Flask API on the Render cloud platform. It includes the

integration of Firebase Services for user authentication and the delivery of low-latency push notifications. The system is designed to operate within a Wi-Fi enabled environment, specifically targeting home-care settings and assisted living facilities. Future iterations may expand this scope to include outdoor tracking via GPS and cellular modules for broader mobility.

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CHAPTER NO. 2

LITERATURE SURVEY

2.1 Literature Survey

Fall detection systems are broadly classified into vision-based and wearable sensor-based approaches. Vision-based methods, such as those by Noury et al. and Rougier et al., use cameras and image processing techniques like motion history analysis. Although accurate, they suffer from privacy issues, limited coverage, and sensitivity to lighting conditions.

Wearable systems, proposed by Bourke et al. and Kangas et al., use accelerometer data with threshold-based detection. These approaches are simple but often produce false alarms during normal activities due to fixed threshold limitations.

Recent work by Bagala et al. and Medrano et al. introduced machine learning techniques that improve detection accuracy by analyzing patterns in sensor data. IoT-based systems like those by Chen et al. enable cloud-based monitoring but depend on smartphones, limiting mobility. Additionally, Igual et al. highlighted the lack of long-term data analysis in existing systems.

2.2 Research Gap

Existing fall detection systems face several limitations. Threshold-based methods result in high false alarms, while vision-based systems raise privacy concerns. Many IoT solutions rely on smartphones or local devices, restricting user mobility. Commercial systems are often expensive and lack open integration. Moreover, most systems do not support long-term data storage or near-fall analysis, which are important for preventive healthcare.

2.3 Proposed Work / Contribution

This project proposes a smart fall detection system addressing these gaps. A machine learning model processes 6-axis sensor data to achieve high accuracy (~95%) and reduce false alarms. An ESP32-based IoT setup enables direct cloud communication without Bluetooth dependency. Real-time alerts are sent within seconds using cloud integration. A Flutter-based app allows caregivers to monitor users remotely. Additionally, all data is stored in Firebase, supporting historical analysis and preventive insights.

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CHAPTER NO. 3

PROPOSED SYSTEM

3.1 System Description

The proposed The proposed FallAlert system is a multi-tier architecture consisting of a sensor node, a cloud-based inference engine, and a mobile interface. The sensor node is worn by the patient and utilizes the MPU6050 to sample motion data at high frequencies. This data is packaged and sent via an HTTP POST request to our Flask server, where the AI model evaluates the risk. If a fall is confirmed, the server communicates with Firebase

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(57) Abstract :

Falls among elderly individuals are a major health concern and often result in severe injuries, delayed medical assistance, and loss of independence. Many existing fall detection systems rely on manual triggers, simple threshold-based sensors, or intrusive camera-based monitoring, which limit their reliability, privacy, and accessibility. The present invention discloses an AI-enabled wearable system for real-time fall detection and automated caregiver alerting. The system is implemented as a lightweight wrist-worn device incorporating a motion sensing module comprising an accelerometer and gyroscope. The wearable device continuously monitors the user's movement and orientation and transmits sensor data wirelessly to a cloud-based processing unit. An artificial intelligence model hosted on the cloud analyzes the incoming motion data in real time to accurately identify fall events while minimizing false alarms caused by normal daily activities. When a fall is detected with sufficient confidence, the system automatically logs the event and sends instant notifications to registered caregivers through a mobile application using cloud messaging services. The proposed system operates autonomously without requiring manual activation, ensuring reliability even when the user is unconscious or unable to seek help. The architecture emphasizes affordability, privacy preservation, and scalability, making it suitable for home care, assisted living facilities, and healthcare monitoring applications. By integrating wearable sensing, artificial intelligence, cloud computing, and mobile communication, the invention provides a comprehensive solution for improving elderly safety and emergency response effectiveness.

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